

Step 1: Reading the Uploaded

```
# استيراد المكتبات
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import folium
```

```
# قراءة البيانات
df = pd.read_csv("FoodDelivery_Saudi_2022-2025.csv")
```

Step 2: Exploring the Dataset

```
print("Shape of data:", df.shape)
```

```
Shape of data: (10000, 7)
```

```
# عرض لاسماء الأعمدة
print(df.columns)
```

```
Index(['Order Number', 'Order Date and Time', 'Order_City', 'Restaurant Type',
       'Total Bill (in Saudi Riyals)', 'Delivery Duration (in minutes)',
       'Customer Rating (from 1 to 5 stars)'],
      dtype='object')
```

```
print("\n أول 5 صفوف من البيانات")
print(df.head())
```

```
أول 5 صفوف من البيانات
Order Number Order Date and Time Order_City Restaurant Type \
0           203   1/18/2025 12:17   Jeddah         Sweets
1           204   5/12/2023 19:40  Buraidah        Barbecue
2           205   5/10/2023 7:47    Hail          Coffee
3           206   4/3/2023 23:39    Abha          Fast Food
4           207   12/21/2024 13:30  Buraidah        Coffee

Total Bill (in Saudi Riyals) Delivery Duration (in minutes) \
0                252.73                15
1                535.70                 7
2                395.11                14
3                698.11                 5
4                499.87                 8

Customer Rating (from 1 to 5 stars)
0                3
1                4
2                4
3                5
4                5
```

```
print("# عرض معلومات عن الأعمدة وأنواع البيانات")
print(df.info())
```

```
# عرض معلومات عن الأعمدة وأنواع البيانات
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Order Number                        10000 non-null  int64
1   Order Date and Time                10000 non-null  object
2   Order_City                         10000 non-null  object
3   Restaurant Type                    10000 non-null  object
4   Total Bill (in Saudi Riyals)        10000 non-null  float64
5   Delivery Duration (in minutes)      10000 non-null  int64
6   Customer Rating (from 1 to 5 stars) 10000 non-null  int64
```

```
dtypes: float64(1), int64(3), object(3)
memory usage: 547.0+ KB
None
```

```
# التحقق من القيم المفقودة
df.isnull().sum()
```

```

      0
Order Number      0
Order Date and Time      0
Order_City         0
Restaurant Type      0
Total Bill (in Saudi Riyals)      0
Delivery Duration (in minutes)      0
Customer Rating (from 1 to 5 stars)      0

```

```
dtype: int64
```

```
print("\n إحصاءات وصفية للأعمدة العددية")
print(df.describe())
```

```

إحصاءات وصفية للأعمدة العددية
Order Number  Total Bill (in Saudi Riyals)  \
count      10000.000000          10000.000000
mean         5202.500000          423.664271
std          2886.895680          222.786720
min           203.000000           36.040000
25%          2702.750000          232.060000
50%          5202.500000          428.385000
75%          7702.250000          616.887500
max         10202.000000          799.980000

Delivery Duration (in minutes)  Customer Rating (from 1 to 5 stars)
count              10000.000000              10000.000000
mean                 11.946700                2.991300
std                   4.625754                1.219333
min                   4.000000                1.000000
25%                   8.000000                2.000000
50%                  12.000000                3.000000
75%                  16.000000                4.000000
max                  20.000000                5.000000

```

```
# حساب القيم الفريدة في كل عمود
df.nunique()
```

```

      0
Order Number      10000
Order Date and Time      9964
Order_City         10
Restaurant Type      12
Total Bill (in Saudi Riyals)      9390
Delivery Duration (in minutes)      17
Customer Rating (from 1 to 5 stars)      5

```

```
dtype: int64
```

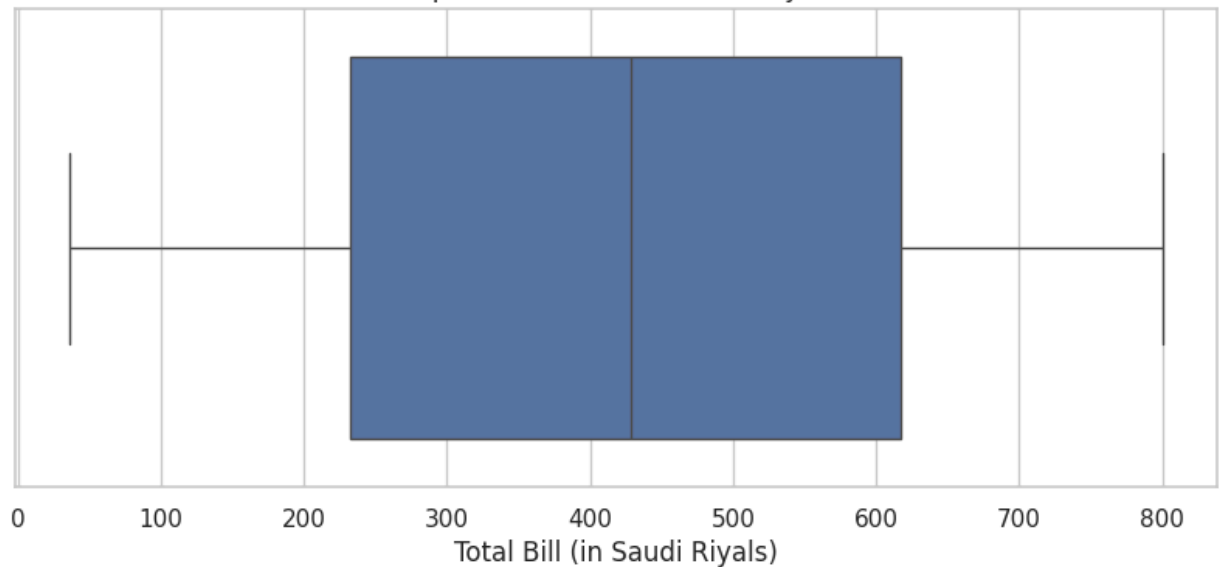
```
# التحقق من الصفوف المكررة
df.duplicated().sum()
```

```
np.int64(0)
```

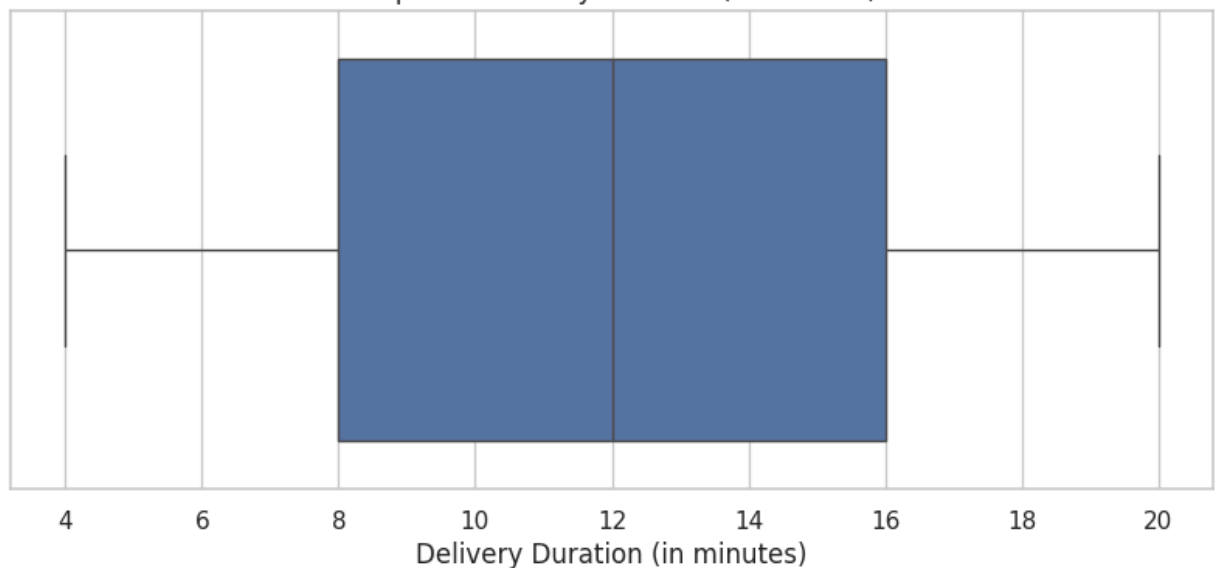
```
# إعداد العرض للرسوم البيانية  
sns.set(style="whitegrid")  
plt.rcParams['figure.figsize'] = (10,4)
```

```
# الأعمدة العددية  
numeric_cols = ['Total Bill (in Saudi Riyals)', 'Delivery Duration (in minutes)', 'Customer Rating (fro  
# رسم بياني لـ Boxplot for outlier  
for col in numeric_cols:  
    plt.figure()  
    sns.boxplot(x=df[col])  
    plt.title(f'Boxplot of {col}')    plt.show()
```

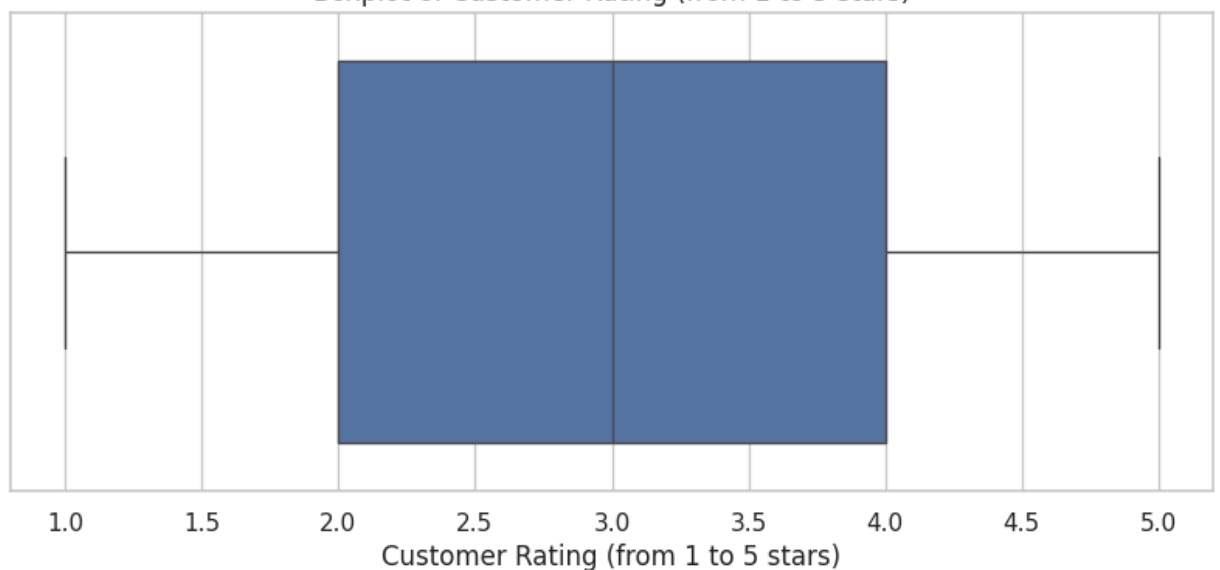
Boxplot of Total Bill (in Saudi Riyals)



Boxplot of Delivery Duration (in minutes)



Boxplot of Customer Rating (from 1 to 5 stars)



```
# باستخدام IQR اكتشاف Outliers
for col in numeric_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
```

```

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]

print(f"\nOutliers in '{col}':")
if outliers.empty:
    print("No outliers detected.")
else:
    print(outliers[['Order Number', col]])

```

Outliers in 'Total Bill (in Saudi Riyals)':
No outliers detected.

Outliers in 'Delivery Duration (in minutes)':
No outliers detected.

Outliers in 'Customer Rating (from 1 to 5 stars)':
No outliers detected.

our data have no missing data or outliers or duplicate rows.

```

# توزيع تقييم العملاء
sns.countplot(x="Customer Rating (from 1 to 5 stars)", data=df, palette="viridis")
plt.title("Customer Rating Distribution")
plt.show()

```

/tmp/ipykernel_28931/2467968111.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` v

```

sns.countplot(x="Customer Rating (from 1 to 5 stars)", data=df, palette="viridis")

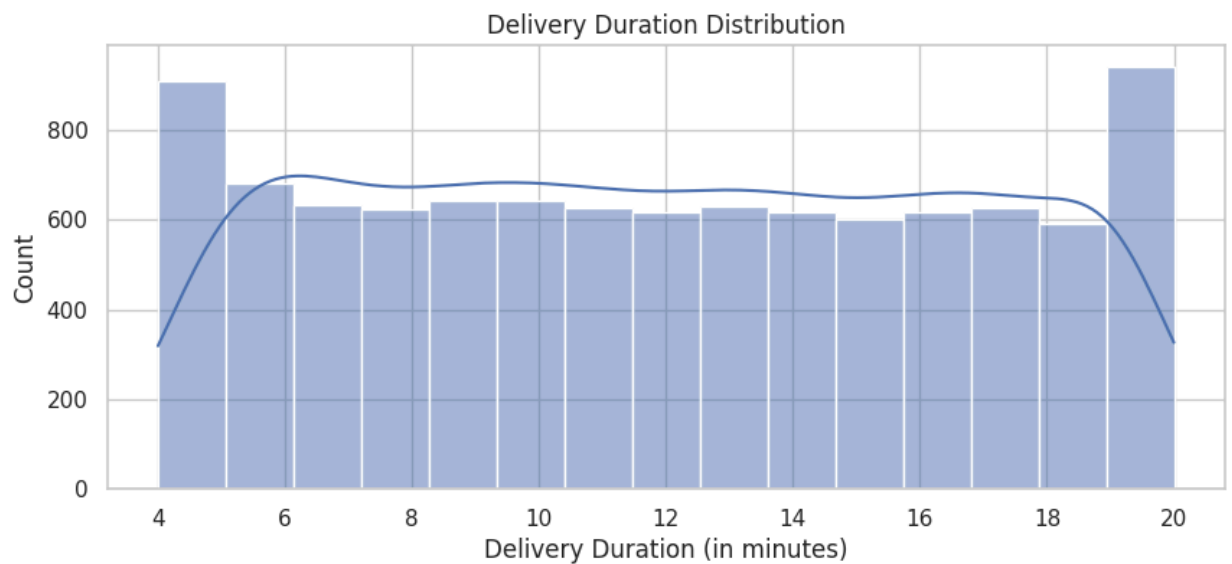
```



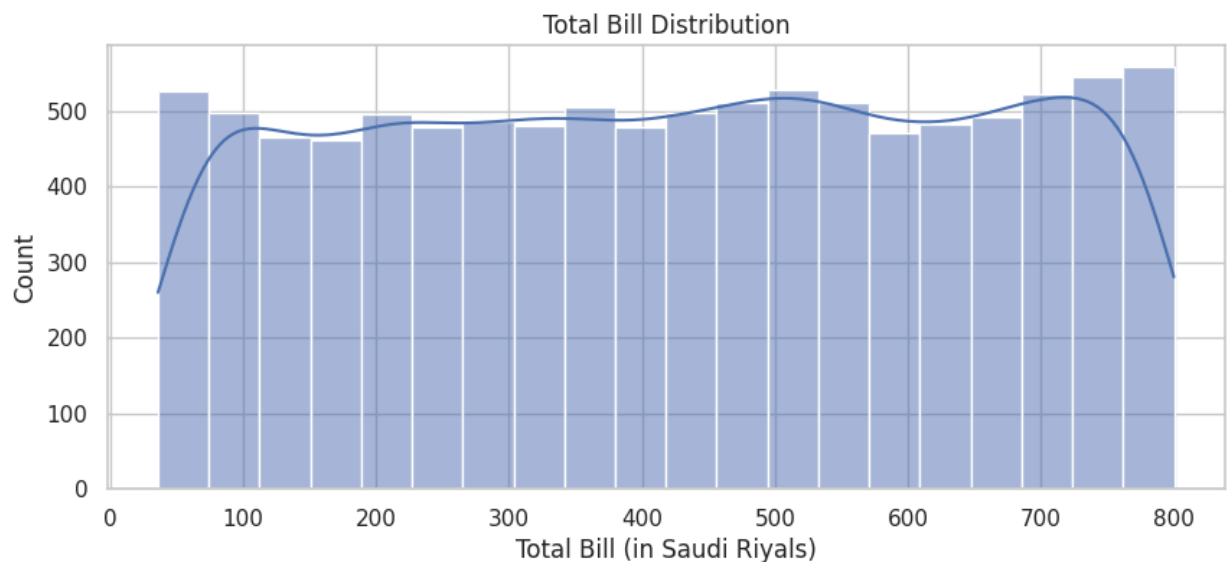
```

# توزيع مدة التوصيل
sns.histplot(df["Delivery Duration (in minutes)"], bins=15, kde=True)
plt.title("Delivery Duration Distribution")
plt.show()

```

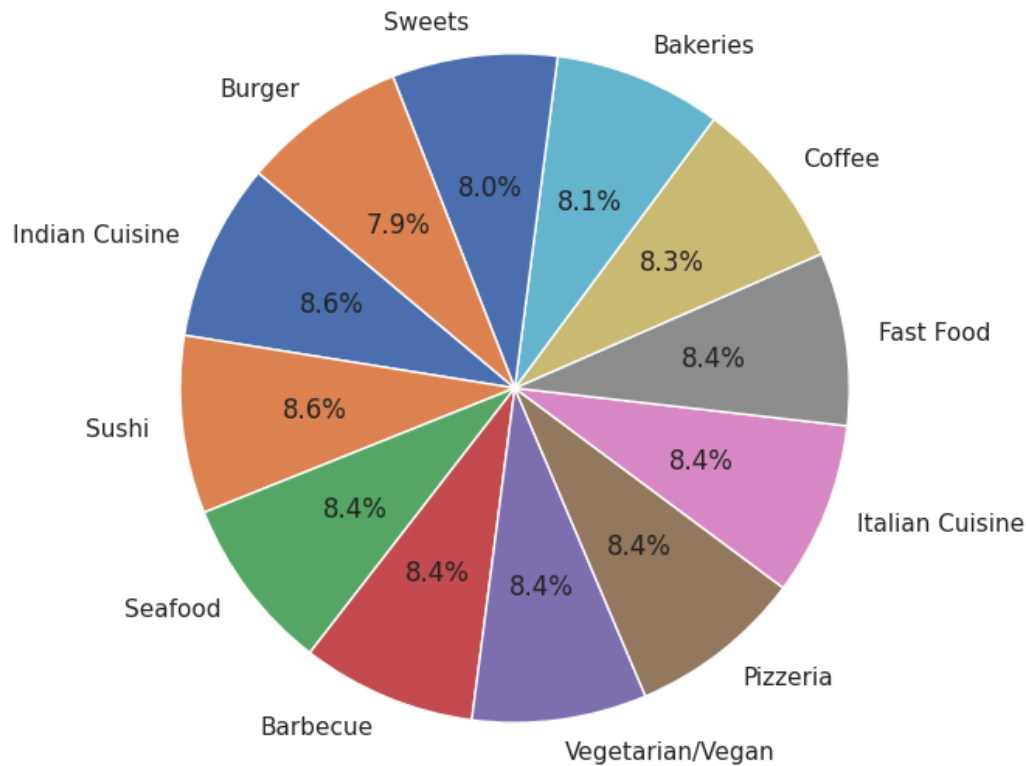


```
# توزيع الفواتير
sns.histplot(df["Total Bill (in Saudi Riyals)"], bins=20, kde=True)
plt.title("Total Bill Distribution")
plt.show()
```



```
# معرفة أكثر نوع مطعم عليه طلبات
orders_by_type = df['Restaurant Type'].value_counts()
plt.figure(figsize=(7,7))
plt.pie(orders_by_type.values, labels=orders_by_type.index, autopct='%1.1f%%', startangle=140)
plt.title("Restaurant Type Orders")
plt.show()
```

Restaurant Type Orders



متوسط التقييم لكل مدينة

```
city_rating = df.groupby("Order_City")["Customer Rating (from 1 to 5 stars)"].mean().sort_values()
city_rating.plot(kind="bar", figsize=(8,4))
plt.title("Average Rating per City")
plt.show()
```

Average Rating per City



العلاقة بين مدة التوصيل والتقييم

```
sns.boxplot(
    x="Customer Rating (from 1 to 5 stars)",
    y="Delivery Duration (in minutes)",
    data=df
)
```

```
plt.title("Delivery Time vs Rating")
plt.show()
```



```
# مدن حسب عدد الطلبات 5
top_cities = df["Order_City"].value_counts().head(5)
print("\n Top 5 Cities by Orders:\n", top_cities)
```

```
Top 5 Cities by Orders:
Order_City
Dammam      1043
Buraidah    1025
Jeddah      1023
Tabuk       1022
Hail        1001
Name: count, dtype: int64
```

```
# أنواع مطاعم حسب متوسط الفاتورة 5
top_restaurants = df.groupby("Restaurant Type")["Total Bill (in Saudi Riyals)"].mean().sort_values(ascending=True)
print("\n Top 5 Restaurant Types by Avg Bill:\n", top_restaurants)
```

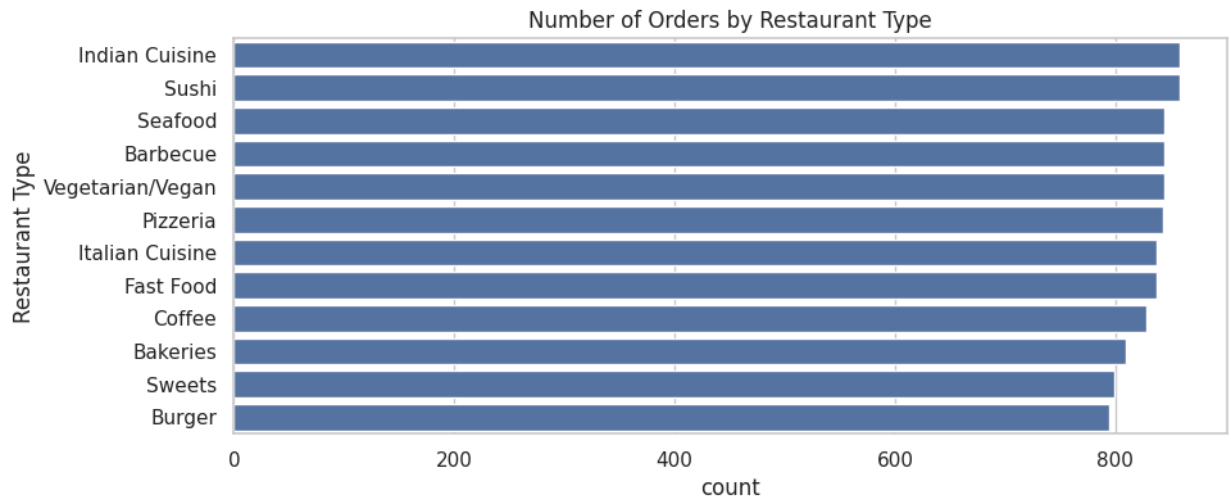
```
Top 5 Restaurant Types by Avg Bill:
Restaurant Type
Vegetarian/Vegan    438.027417
Bakeries            437.364778
Indian Cuisine      430.289371
Pizzeria            427.713084
Italian Cuisine     425.992876
Name: Total Bill (in Saudi Riyals), dtype: float64
```

```
# حسب التقييم (مدينة + نوع مطعم) 5
city_rest_rating = df.groupby(["Order_City", "Restaurant Type"])["Customer Rating (from 1 to 5 stars)"]
print("\n Top 5 (City + Restaurant Type) by Avg Rating:\n", city_rest_rating)
```

```
Top 5 (City + Restaurant Type) by Avg Rating:
Order_City Restaurant Type
Dammam      Barbecue      3.297619
Khobar      Coffee        3.246575
Buraidah    Pizzeria       3.243590
Dammam      Burger        3.236111
Hail        Bakeries       3.232877
Name: Customer Rating (from 1 to 5 stars), dtype: float64
```

```
# أنواع المطاعم
plt.figure()
sns.countplot(data=df, y='Restaurant Type', order=df['Restaurant Type'].value_counts().index)
```

```
plt.title('Number of Orders by Restaurant Type')
```



Step 3: Data Cleaning & Preprocessing

دالة التصنيف

```
def classify_food(x):
    if x in ["Burger", "Fast Food", "Barbecue"]:
        return "American"
    elif x in ["Italian Cuisine", "Pizzeria"]:
        return "Italian"
    elif x in ["Indian Cuisine", "Sushi"]:
        return "Asian"
    elif x in ["Bakeries", "Sweets"]:
        return "Desserts & Bakery"
    elif x in ["Coffee", "Seafood", "Vegetarian/Vegan"]:
        return x # يترك الاسم كما هو
    else:
        return x # أي اسم آخر يبقى كما هو
```

تطبيق وحفظ التصنيف على العامود

```
df["Restaurant Type"] = df["Restaurant Type"].apply(classify_food)
```

from google.colab import files

بدون أي تغييرات إضافية Excel نحفظ الملف النهائي بصيغة
df.to_excel("file_final.xlsx", index=False)

تنزيل الملف على الجهاز

```
files.download("file_final.xlsx")
```